**5.11** Solution: (a) The loop with normal having spherical angles $\theta_0$ and $\phi_0$ can be viewed as taking a loop lying in the $x$-$y$ plane by first rotating $\theta_0$ around $y$-axis, and $\phi_0$ around $z$-axis. Then any point or direction originally in the $x$-$y$ plane can be determined by multiplying the vector with the rotation matrix which is given by

$$T = \begin{pmatrix} \cos\phi_0 & -\sin\phi_0 & 0 \\ \sin\phi_0 & \cos\phi_0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta_0 & 0 & \sin\theta_0 \\ 0 & 1 & 0 \\ -\sin\theta_0 & 0 & \cos\theta_0 \end{pmatrix}.$$

Point $a(\cos\phi, \sin\phi, 0)$ becomes

$$\mathbf{u}(\phi) = a \begin{pmatrix} \cos\phi\cos\theta_0\cos\phi_0 - \sin\phi\sin\phi_0 \\ \cos\phi\cos\theta_0\sin\phi_0 + \sin\phi\cos\phi_0 \\ -\cos\phi\sin\theta_0 \end{pmatrix} = a \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix},$$

and direction $(-\sin\phi, \cos\phi, 0)$ becomes

$$\mathbf{v}(\phi) = \begin{pmatrix} -\sin\phi\cos\theta_0\cos\phi_0 - \cos\phi\sin\phi_0 \\ -\sin\phi\cos\theta_0\sin\phi_0 + \cos\phi\cos\phi_0 \\ \sin\phi\sin\theta_0 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}.$$

Now, the we can calculate the force as

$$\begin{aligned}
\mathbf{F} &= \oint I\, d\ell \times \mathbf{B}(\mathbf{u}) = I \int_0^{2\pi} d\ell \times \mathbf{B}(\mathbf{u}(\phi))\, d\phi = Ia \int_0^{2\pi} \mathbf{v} \times \mathbf{B}(\mathbf{u}(\phi))\, d\phi. \\
&= IaB_0 \int_0^{2\pi} \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ v_1 & v_2 & v_3 \\ 1+\beta a u_2 & 1+\beta a u_1 & 0 \end{vmatrix} d\phi \\
&= IaB_0 \int_0^{2\pi} \Big[ -(1+\beta a u_1) v_3\, \hat{\mathbf{i}} + (1+\beta a u_2) v_3\, \hat{\mathbf{j}} + \big( (1+\beta a u_1) v_1 - (1+\beta a u_2) v_2 \big) \hat{\mathbf{k}} \Big] d\phi.
\end{aligned}$$

First order term in $\cos\phi$ and $\sin\phi$ will drop out

$$\mathbf{F} = \beta I a^2 B_0 \int_0^{2\pi} \left( -u_1 v_3\, \hat{\mathbf{i}} + u_2 v_3\, \hat{\mathbf{j}} + (u_1 v_1 - u_2 v_2) \hat{\mathbf{k}} \right) d\phi.$$

Further, $\sin\phi\cos\phi$ terms will also drop out

$$\begin{aligned}
\mathbf{F} &= \beta I a^2 B_0 \int_0^{2\pi} \begin{pmatrix} \sin^2\phi\, \sin\theta_0 \sin\phi_0 \\ \sin^2\phi\, \sin\theta_0\cos\phi_0 \\ \cos^2\phi\cos\theta_0\sin\theta_0 + \cos^2\phi\cos\theta_0\sin\theta_0 \end{pmatrix} d\phi \\
&= \beta I \pi a^2 B_0 \left( \sin\theta_0 \sin\phi_0\, \hat{\mathbf{i}} + \sin\theta_0\cos\phi_0\, \hat{\mathbf{j}} \right).
\end{aligned}$$

After rotation, $\hat{\mathbf{z}}$ direction becomes

$$(\sin\theta_0\cos\phi_0, \sin\theta_0\sin\phi_0, \cos\theta_0).$$

Then

$$\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B}) = I\pi a^2 \nabla(\hat{\mathbf{n}}\cdot\mathbf{B})$$

$$
\begin{aligned}
&= I\pi a^2 B_0 \nabla\Big((1+\beta y)\sin\theta_0\cos\phi_0 + (1+\beta x)\sin\theta_0\sin\phi_0\Big) \\
&= \beta I\pi a^2 B\left(\sin\theta_0\sin\phi_0\,\hat{\mathbf{i}} + \sin\theta_0\cos\phi_0\,\hat{\mathbf{j}}\right)
\end{aligned}
$$

which agrees with exact calculation.

(b) The torque can be directly calculated as

$$
\begin{aligned}
\mathbf{N} &= \mathbf{m}\times\mathbf{B}(0) = I\pi a^2\,\hat{\mathbf{n}}\times\mathbf{B}(0) \\
&= I\pi a^2 B\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \sin\theta_0\cos\phi_0 & \sin\theta_0\sin\phi_0 & \cos\theta_0 \\ 1 & 1 & 0 \end{vmatrix} \\
&= I\pi a^2 B\left(-\cos\theta_0\,\hat{\mathbf{i}} + \cos\theta_0\,\hat{\mathbf{j}} + \sin\theta_0(\cos\phi_0 - \sin\phi_0)\hat{\mathbf{k}}\right).
\end{aligned}
$$